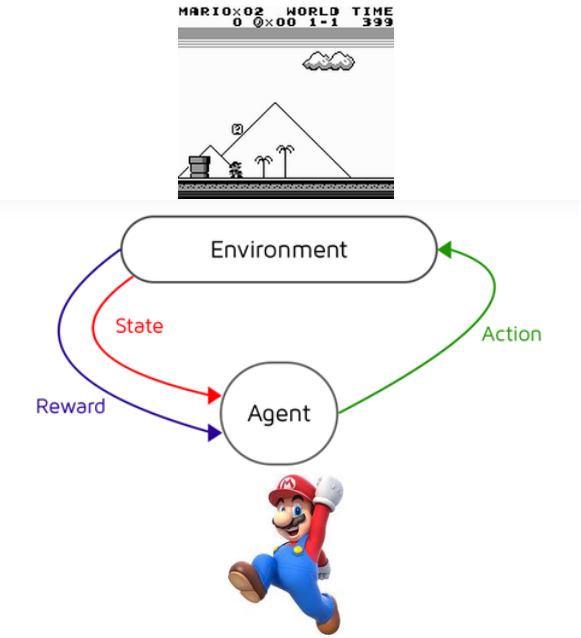
Reinforcement Learning

Fundamentals of Artificial Intelligence

Sequential Decision Making

reinforcement learning (RL):

formalization of sequential decision making (action policy) of software agent interacting with environment



corresponds to search for best (or rather good) action policy to reach a given goal (e.g., win a game)

using learning from examples (data) to guide the search

RL usually more difficult (e.g., non-differentiable as a whole) than supervised learning (which can be seen as “generalized optimization”, often of proxy metric)

Main Elements of RL

goal: find action policy maximizing reward from environment

action policy

- exploration-exploitation trade-off
- e.g., epsilon-greedy: random exploration at small fraction of the time

feedback from environment

- goal-directed, no supervision
- scalar reward signal, cumulative and delayed (credit assignment problem)

Optional Elements of RL

value functions for states or actions: improve efficiency of search in vast action policy space (alternative: direct policy search)

model of environment: (model-free) learning from trial-and-error or (model-based) planning

can be used in different ways:

- simulate experience from model (for learning)
- decision-time planning (e.g., heuristic search or model predictive control)

Markov Decision Process (MDP)

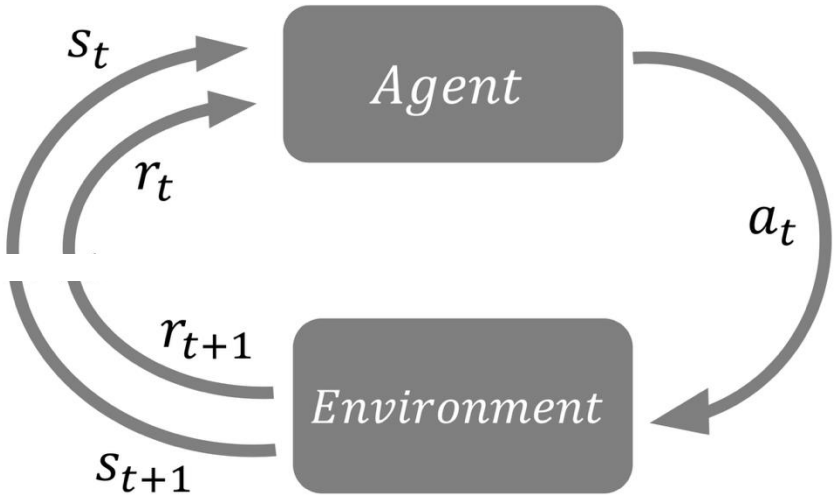
idea: current state includes all information about past

transition probabilities between states describe dynamics of given MDP

action policy: mapping from states to probabilities for selection of different actions

States, Actions, and Rewards

transition probabilities (model of environment): $p(s_{t+1}, r_{t+1} | s_t, a_t)$



reward hypothesis:

- reward as scalar signal
- goal: maximization of expected cumulative sum of received rewards

State and Action Values

state/action value: total amount of expected future reward starting from given state/action (usually with discounting of later steps)

→ indicating long-term desirability of states/actions

main motivation: improve efficiency of search in policy space

(for comparison: evolutionary methods search directly by evaluating entire policies)

State-Value Function

(needed for all states)

on-policy

return

discount rate

$$v_\pi(s_t) = E_\pi\left[\overbrace{\sum_{k=o}^{\infty} \gamma^k r_{t+k+1}}^{\text{return}} \,|s_t\right] = E_\pi[r_{t+1} + \gamma v_\pi(s_{t+1})|s_t]$$

$$= \sum_{a_t} \pi(a_t|s_t) \sum_{s'_{t+1}, r_{t+1}} p(s'_{t+1}, r_{t+1}|s_t, a_t)[r_{t+1} + \gamma v_\pi(s'_{t+1})]$$

policy: probabilitiy to take specific action being in a given state

transition probability (depending on environment) from state s_t to state s'_{t+1} for a given action

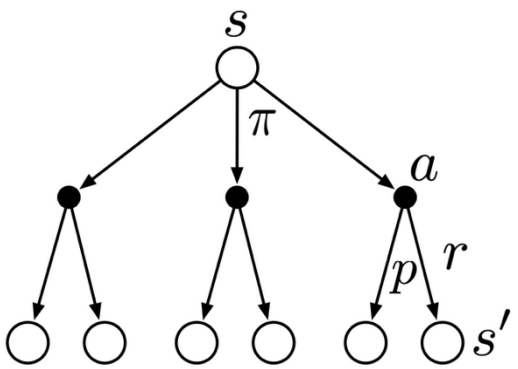
(sweep through entire state space)

Bellman (expectation) equation: recursion

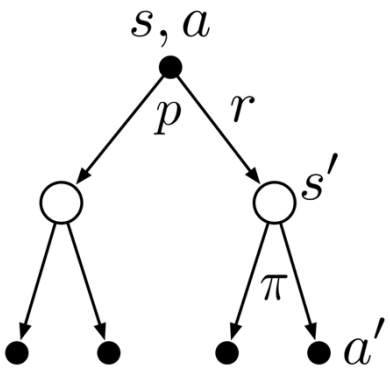
Action-Value Function

$$q_\pi(s_t, a_t) = E_\pi\left[\sum_{k=o}^{\infty} \gamma^k r_{t+k+1} \,|s_t, a_t\right] = E_\pi[r_{t+1} + \gamma q_\pi(s_{t+1}, a_{t+1})|s_t, a_t]$$

$$= \sum_{s'_{t+1}, r_{t+1}} p(s'_{t+1}, r_{t+1}|s_t, a_t)\left[r_{t+1} + \gamma \sum_{a'_{t+1}} \pi(a'_{t+1}|s'_{t+1})\, q_\pi(s'_{t+1}, a'_{t+1})\right]$$



Backup diagram for v_π



q_π backup diagram

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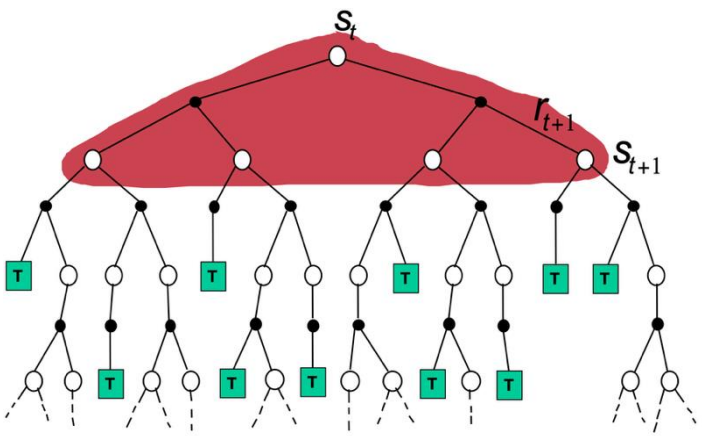
Bootstrapping and Sampling

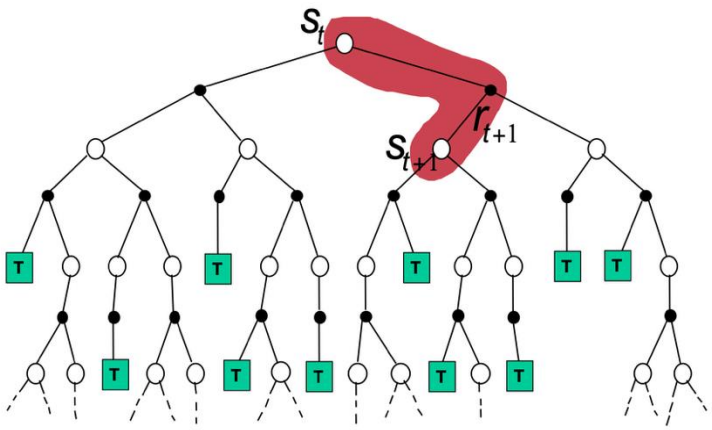
bootstrapping: update estimates of state values based on estimates of values of successor states
sampling: experience of sample sequences (no need for complete knowledge of environment)

Dynamic Programming



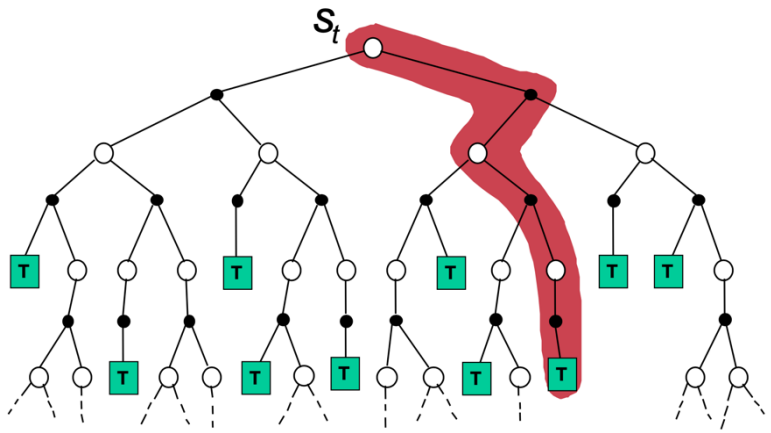
- bootstrapping
- no sampling → model-based (transition probabilities needed)

Temporal Difference (TD) Learning



- bootstrapping
- sampling → model-free

Monte Carlo (MC)



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- no bootstrapping
- sampling → model-free

Sampling Update Rule

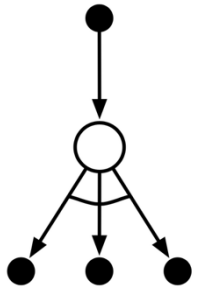
$$\boxed{NewEstimate \leftarrow OldEstimate \; + \; StepSize \left[Target - OldEstimate \right]}$$

MC: $v(s_t) \leftarrow v(s_t) + \eta[\sum_{k=o}^{\infty} \gamma^k r_{t+k+1} - v(s_t)]$

TD: $v(s_t) \leftarrow v(s_t) + \eta[r_{t+1} + \gamma v(s_{t+1}) - v(s_t)]$

bootstrapping

Q-Learning



approximation of action-value function:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \eta \left[r_{t+1} + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t) \right]$$

behavior policy (used to collect data) can differ from target policy (greedy) → off-policy

Q stands for quality function: estimate of true action-value function

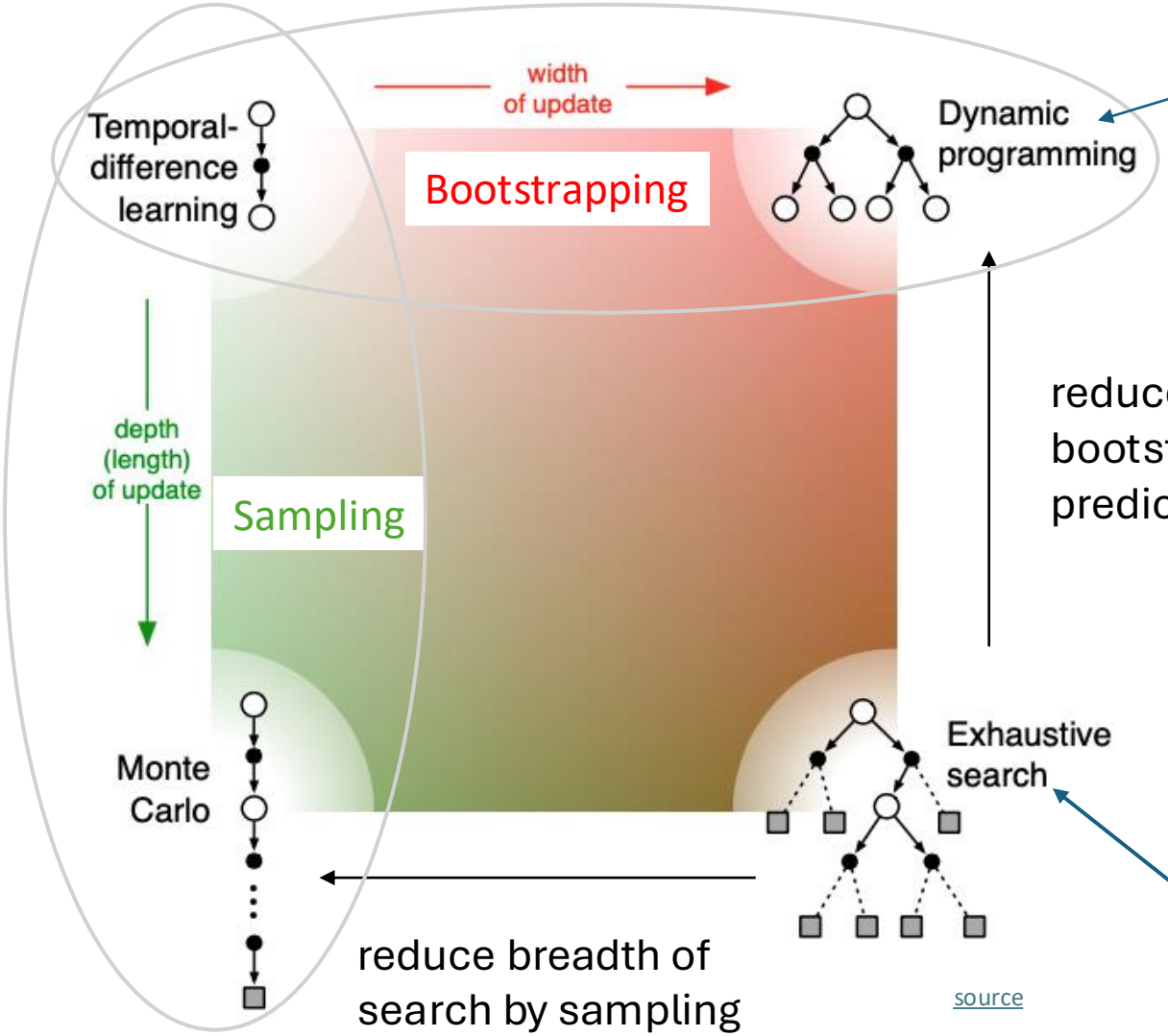
control strategy: choose action with highest Q-value (exploitation)

Update Characteristics

one-step TD

n-step TD

MC



model of environment needed

reduce depth of search by bootstrapping with value function predicting future outcome

reduce breadth of search by sampling actions from policy

very inefficient

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Limitation of Tabular Methods

tabular methods (calculating values for each state/action) simply memorize observed data

problem with tabular solution methods in practice: large state/action spaces (kind of curse of dimensionality)

→ need for generalization: supervised learning to the rescue

- non-linear function approximation over state/action space
- nowadays often deep learning methods → deep RL

Approximate Solution Methods

online setting: learning while interacting with environment
offline setting: learning from limited data without further interaction with environment

state/action values as parametrized function (instead of table)

- variables/features describing different states
- parameters (e.g., weights in neural network) to be learned

objective function for supervised learning (e.g., squared error loss):

$$\mathcal{L}(\boldsymbol{w}) = \sum_{s} \left(v(s) - \hat{v}(s; \boldsymbol{w}) \right)^2$$

parameters/weights to be optimized

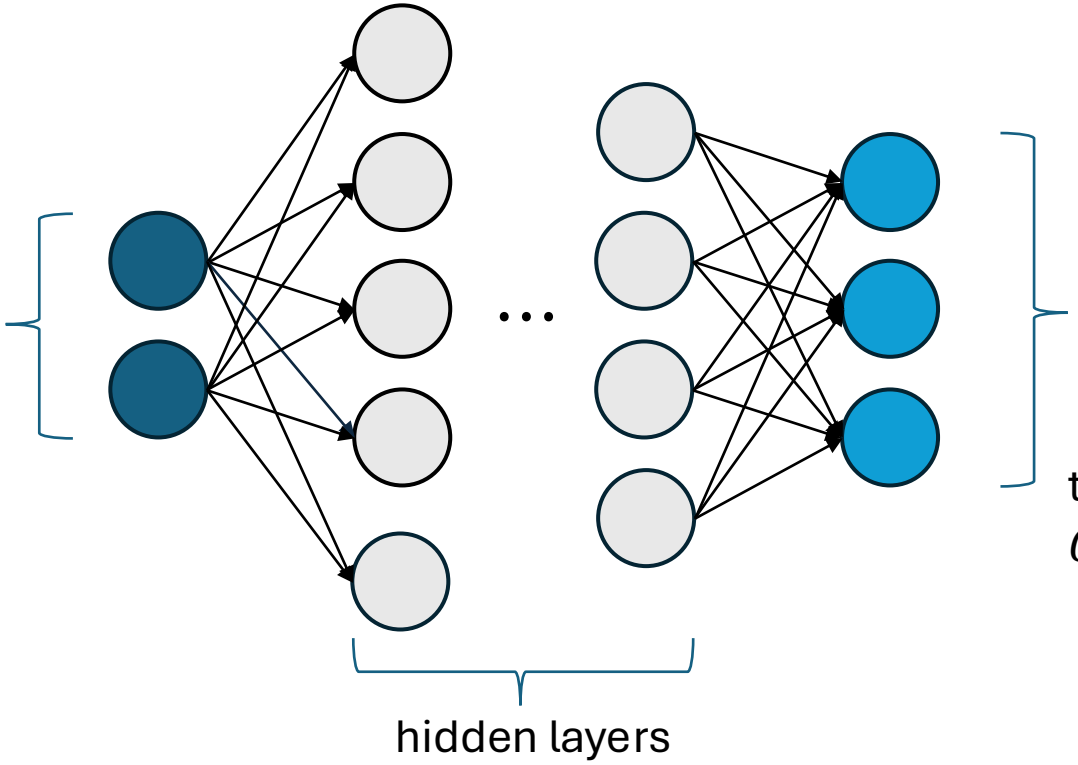
→ RL problem expressed in supervised learning setup

but $v(s)$ still calculated via RL methods (e.g., bootstrapping)

Deep Q-Network (DQN)

idea: approximation of tabular Q-function by neural network

values describing the state



Q-values for the different actions

target:
$Q(s_t, a_t) = r_{t+1} + \gamma \max_a Q(s_{t+1}, a_{t+1})$

Key Components of DQN

tricks for training stabilization:

experience replay:

training samples (or mini batches) drawn at random from previously stored experiences → remove correlations in observation sequence

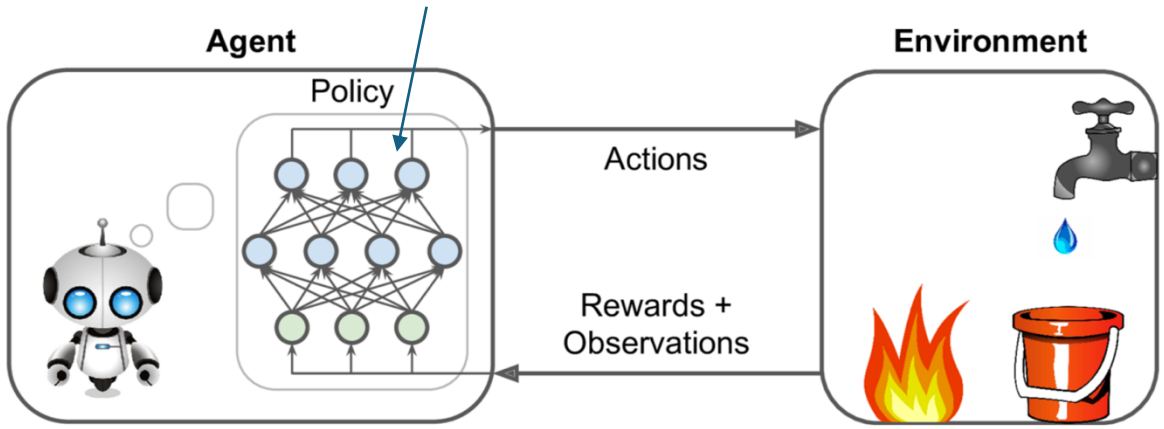
separate target network:

copy of Q-network, but only periodically updated → reduce target correlations due to bootstrapping

Direct Policy Search: Policy Gradient Method

neural network with actions as output/target: policy network

action probabilities



observed returns
(MC sampling)

$$\text{loss: } -\log P(a_t|s_t;\boldsymbol{w}) \cdot (r_{t+1} + \gamma r_{t+2} + \cdots)$$

→ policy gradients

gradient ascent
→ actions with higher rewards

choose action according to probability
→ implicit exploration, on-policy method

prominent method: Proximal Policy Optimization (PPO)

Actor-Critic Methods

interpretation: $-\log P(a_t|s_t;\boldsymbol{w}) \cdot \underbrace{(r_{t+1} + \gamma r_{t+2} + \cdots)}_{\text{Critic}}$

Actor

no change in gradient when subtracting action-independent baseline:

$$\boxed{\text{loss: } -\log P(a_t|s_t;\boldsymbol{w}) \cdot \underbrace{[(r_{t+1} + \gamma r_{t+2} + \cdots) - B]}_{\text{advantage}}}$$

advantage (extra reward for specific action compared to average action in given state)

possible baselines:

- average discounted return
- value function (MC → TD): hybrid between policy-based and value-based methods

→ reduction of variance

AlphaGo, AlphaGo Zero

Famous Example of Deep RL: AlphaGo

Monte Carlo tree search (heuristic, lookahead search) for move (i.e., action) selection

→ decision-time planning (advantage: focus on current state rather than full state space)

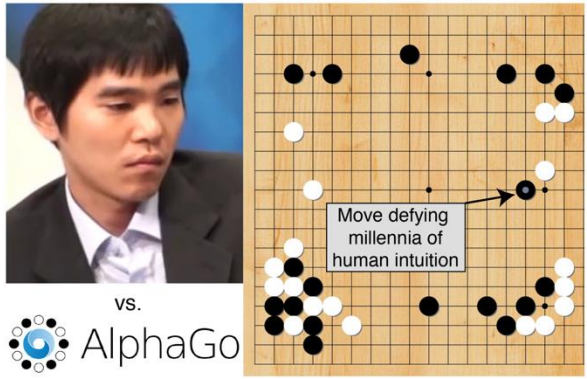
guided by deep convolutional neural networks for both value function and policy estimation

→ improving search efficiency

reduce depth of search tree by evaluating positions with **value function** (predicting outcome from given position → **bootstrapping**)

reduce breath of search tree by **sampling** actions using **policy network** (probability distribution over possible moves in given position)

famous move 37:



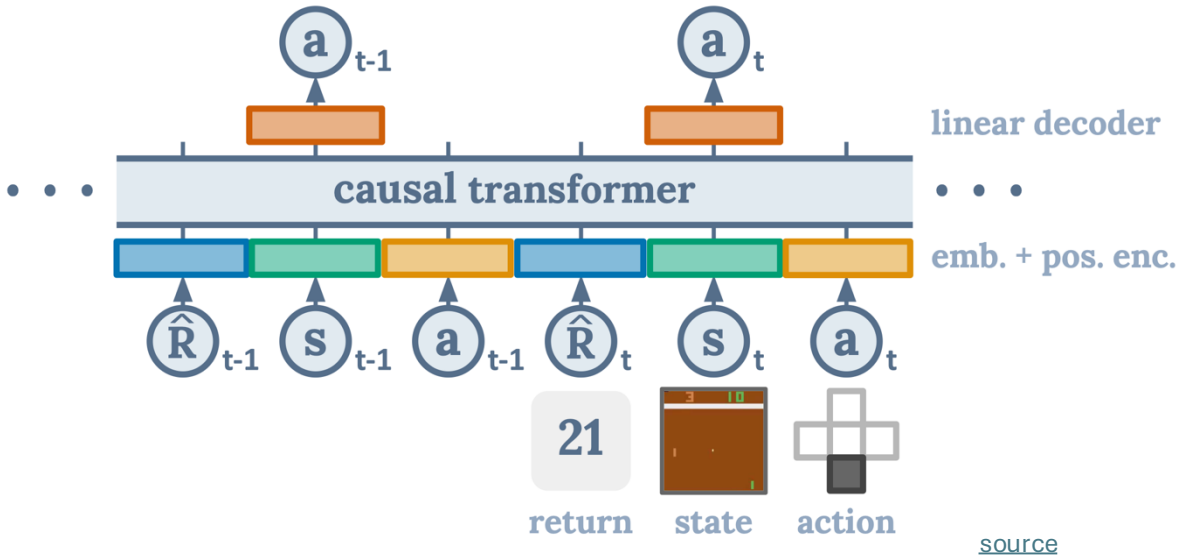
from Max Tegmark

Sequence Modeling for Decisions/Actions

generative: transformer decoder architecture to autoregressively model trajectories

credit assignment directly via self-attention: implicitly forming state-return associations via similarity of query and key vectors (maximizing the dot product)

desired return tokens as prompt for action generation



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Decision Transformer: conditioning on desired return, past states and actions to generate future actions